Tristan Toshiharu Endo

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Education

University of California, San Diego Expected June 2027
B.S. Mathematics–Computer Science *La Jolla, CA*

- **Graduate Coursework:** Machine Learning (DSC 240), Probability Theory (MATH 280A), Applied Statistics (MATH 282A)
- **Relevant Coursework:** Mathematical Methods in Data Science I–III, Optimization for Data Science, Mathematical Statistics I–II, Data Analysis and Inference, Probability and Stochastic Processes, Combinatorics, Mathematics of Finance, Data Structures (in progress)

Research

UC San Diego, Undergraduate Researcher Feb 2026 – Present
Advisor: Raymond O. Chou Collaborator: Alexander Ryan *La Jolla, CA*

- **Springer Fibers in Algebraic Geometry:** Built a Python/SymPy pipeline that enumerates the cells of type A Springer fibers, computes their Plücker coordinates, and searches for witness curves certifying closure relations between cells
- Extended the pipeline from two-row Jordan types to three- and four-row types, where existing combinatorial theory does not apply; computed closure data for nearly 100 Jordan types, about 20,000 cells, and over 500,000 relations
- Fixed a span-check bug that stalled larger cases, recovering degree-4 and -6 witnesses for $(4, 4)$ and $(5, 5)$; an exact ideal-based check confirmed every found relation and revealed missed intersections in 68 of 210 closures

Experience

Tekly Studio, Quantitative Research & AI Engineering Lab Jan 2026 – Aug 2026
Quantitative Researcher / AI Engineering Intern *San Francisco, CA*

- Built and backtested an XGBoost momentum strategy for cryptocurrency markets in Python and carried it from research into production, evaluating performance via Sharpe ratio and drawdown
- Prevented look-ahead leakage with purged k-fold cross-validation and walk-forward backtesting, and engineered pipelines for preprocessing, simulation, and model evaluation

Think Twice Legal Apr 2025 – Jul 2026
Audio Engineer / Analyst *San Bruno, CA*

- Optimized Demucs-based source separation models via hyperparameter tuning, improving reconstruction fidelity for forensic audio analysis used in legal proceedings
- Designed FFT-based noise reduction and normalization pipelines for consistent model input across varying sample rates and noise conditions

Selected Projects

Neural Network Gaussian Processes | *Python* Jan 2026 – Mar 2026

- Course project in MATH 216B (Prof. Alexander Cloninger). Implemented closed-form conjugate kernels for ReLU, erf, and tanh networks and showed how their correlations degenerate or decorrelate with depth in the infinite-width limit; tested finite-width simulations at fixed width against these predictions

Springer Fiber Visualizer | *JavaScript, HTML Canvas* Feb 2026 – Present

- Built an interactive 3D web viewer for the closure data, with nested intersections, a searchable relation matrix, and in-browser exact checks of each witness curve

Stochastic Simulation of Brownian Motion & OU Processes | *Python* Sep 2025 – Dec 2025

- Simulated fractional Brownian motion across varying Hurst parameters to characterize long-range dependence; quantified empirical extrema distributions via large-scale Monte Carlo simulation

Technical Skills

Languages & Tools: Python, C, C++, Java, SQL, R, MATLAB, HTML/JavaScript, Git, Linux, Jupyter, LaTeX
ML & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, SymPy, FFT/signal processing
Methods: Model evaluation and validation, time-series modeling, stochastic processes, statistical inference